WORKED SOLUTION

STEP II 2018, Question 4

Pure • Trigonometric Equations

STEP II 2018, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q4

The key idea

The given identity turns a sum of cosines into a product, and a product is what we want when solving an equation equal to zero. In (i), pairing the outer terms and the inner terms produces a common factor at once. In (ii) and (iii) the useful move is to express everything through the half-sum and half-difference of the angles, after which (ii) factorises and (iii) becomes a sum of two squares equal to zero.

Solution

Part (i)

Pair the first and last terms, and the two middle terms. By the identity,

$$\cos x + \cos 4x = 2\cos\tfrac{5x}{2}\cos\tfrac{3x}{2},$$
$$3(\cos 2x + \cos 3x) = 6\cos\tfrac{5x}{2}\cos\tfrac{x}{2}.$$

(We may write $\cos(-\frac{3x}{2}) = \cos\frac{3x}{2}$, since cosine is even.) So the left-hand side is

$$2\cos\tfrac{5x}{2}\left(\cos\tfrac{3x}{2} + 3\cos\tfrac{x}{2}\right).$$

With $c = \cos\frac{x}{2}$, the triple-angle formula $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ gives $\cos\frac{3x}{2} = 4c^3 - 3c$, so the bracket is just $4c^3$. Hence the equation is

$$8\cos\tfrac{5x}{2}\cos^3\tfrac{x}{2} = 0.$$

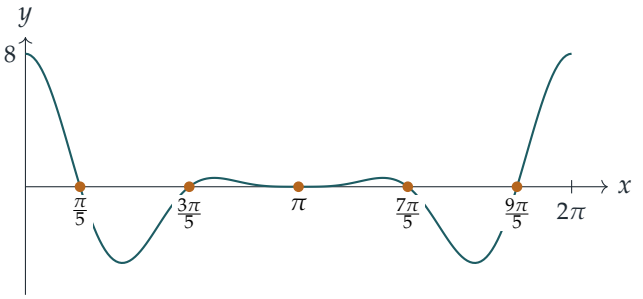
First factor. $\cos\frac{5x}{2} = 0$ means $\frac{5x}{2} = \frac{\pi}{2} + k\pi$ for an integer k, i.e. $x = \frac{(2k+1)\pi}{5}$. For $0 \le x \le 2\pi$ we need $0 \le 2k+1 \le 10$, so $k = 0, 1, 2, 3, 4$.

Second factor. $\cos\frac{x}{2} = 0$ with $0 \le \frac{x}{2} \le \pi$ gives only $\frac{x}{2} = \frac{\pi}{2}$, i.e. $x = \pi$, which is already in the list.

Therefore the solutions are

$$x = \frac{\pi}{5},\ \frac{3\pi}{5},\ \pi,\ \frac{7\pi}{5},\ \frac{9\pi}{5}.$$

The graph of the left-hand side shows them; at $x = \pi$ both $\cos\frac{5x}{2}$ and $\cos^3\frac{x}{2}$ vanish, so the curve only touches the axis there rather than crossing it.



Part (ii)

Using the identity on the first two terms, and $\cos 2x = 2\cos^2 x - 1$,

$$\cos(x+y) + \cos(x-y) - \cos 2x = 2\cos x \cos y - 2\cos^2 x + 1.$$

Setting this equal to 1 gives

$$2\cos x\,(\cos y - \cos x) = 0,$$

so $\cos x = 0$ or $\cos y = \cos x$.

- If $\cos x = 0$ with $0 \le x \le \pi$, then $x = \frac{\pi}{2}$ (and then any y works).
- If $\cos y = \cos x$, then $x = y$, because cosine is strictly decreasing on $[0, \pi]$ and so takes each value there only once.

So either $x = y$ or $x = \frac{\pi}{2}$, as required.

Part (iii)

Let $s = \frac{1}{2}(x+y)$ and $d = \frac{1}{2}(x-y)$. By the identity $\cos x + \cos y = 2\cos s \cos d$, and $\cos(x+y) = \cos 2s = 2\cos^2 s - 1$. The equation becomes

$$2\cos s\cos d - 2\cos^2 s + 1 = \tfrac{3}{2} \iff 4\cos^2 s - 4\cos s\cos d + 1 = 0.$$

Ask yourself One equation, two unknowns: how can it pin down both x and y?

Complete the square in $\cos s$. Since $1 = \cos^2 d + \sin^2 d$,

$$4\cos^2 s - 4\cos s\cos d + \cos^2 d + \sin^2 d = (2\cos s - \cos d)^2 + \sin^2 d = 0.$$

A sum of two real squares is zero only if both are zero, which gives two equations:

$$\sin d = 0 \qquad \text{and} \qquad 2\cos s = \cos d.$$

Since $x, y \in [0, \pi]$, we have $-\frac{\pi}{2} \le d \le \frac{\pi}{2}$, and the only zero of sin there is $d = 0$. So $x = y$, and then $s = x$ and $\cos d = 1$, giving $\cos x = \frac{1}{2}$. With $0 \le x \le \pi$ this means $x = \frac{\pi}{3}$.

So the only solution is

$$x = y = \frac{\pi}{3}.$$

Indeed, $\cos\frac{\pi}{3} + \cos\frac{\pi}{3} - \cos\frac{2\pi}{3} = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$.

The same algebra shows more: for all x, y,

$$\tfrac{3}{2} - \big(\cos x + \cos y - \cos(x+y)\big) = \tfrac{1}{2}\big[(2\cos s - \cos d)^2 + \sin^2 d\big] \ge 0,$$

so $\frac{3}{2}$ is the largest value the left-hand side can take, and the equation asks exactly where that maximum is reached.

Checks

- In (i), at $x = \frac{\pi}{5}$: $\cos\frac{\pi}{5} + \cos\frac{4\pi}{5} = 0$ and $\cos\frac{2\pi}{5} + \cos\frac{3\pi}{5} = 0$, since $\cos(\pi - \theta) = -\cos\theta$. At $x = 0$ the left-hand side is $1 + 3 + 3 + 1 = 8$, matching $8\cos 0\cos^3 0$.
- A computer search on a fine grid over $[0, 2\pi]$ found exactly the five zeros above, and a grid over $[0, \pi]^2$ found the maximum of $\cos x + \cos y - \cos(x+y)$ to be 1.5, attained only at $x = y \approx 1.0472 = \frac{\pi}{3}$.
- In (ii), $x = \frac{\pi}{2}$ gives $\cos(\frac{\pi}{2} + y) + \cos(\frac{\pi}{2} - y) - \cos\pi = -\sin y + \sin y + 1 = 1$ for every y.

Common slips

- Listing $x = \pi$ twice, or missing it: it comes from both factors in (i) but is one solution.
- In (ii), dividing by $\cos x$ and so losing $x = \frac{\pi}{2}$.
- In (iii), stopping at a quadratic in $\cos s$ without using the discriminant (or the squares) to force $\cos^2 d = 1$.

Beyond STEP

The coefficients $1, 3, 3, 1$ in (i) are no accident: the left-hand side is the real part of $e^{ix}(1 + e^{ix})^3$, and since $1 + e^{ix} = 2\cos\frac{x}{2}\, e^{ix/2}$ this is $8\cos^3\frac{x}{2}\cos\frac{5x}{2}$ in one line. Part (iii) is the classical inequality $\cos A + \cos B + \cos C \le \frac{3}{2}$ for the angles of a triangle in disguise (put $A = x$, $B = y$, $C = \pi - x - y$), with equality only for the equilateral triangle; maximising a function of two variables like this is a standard first-year topic in multivariable calculus. Needham's *Visual Complex Analysis* shows how complex exponentials make trigonometric identities like these almost automatic.

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